CogniNet-6G: An AI-Native Closed-Loop Digital Twin Framework for Predictive Semantic Network Orchestration in Private 6G Industrial Robot Networks

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Abstract To operate effectively on a fully automated shop floor, high bandwidth inspection feeds and high reliability, low latency operation are required. Control of functional safety also involves sub-millisecond control. Line of sight disruptions are common with physical objects like overhead gantry cranes, tall storage racks, and moving steel containers in high-frequency 28 GHz mmWave private cellular links, causing sudden signal fluctuations. In conventional network slices schedulers, quotas for Physical Resource Blocks (PRB) for network slices are dynamically adjusted after the queues have built up and packets have been dropped due to link quality. This reaction lag often exceeds communication keep alive periods, leading to a mechanical emergency brake that brings the fleet of mobile robots to a halt and halts line production, a situation where the robots are traveling at speeds between 1.5 to 2.0 m/s. In this work, we introduce CogniNet-6G, a closed-loop digital twin framework for predicting wireless link degradation and preventing the impact of delays on safety-critical traffic. A 2-layer bidirectional Long Short-Term Memory (Bi-LSTM) network is used for slice demand prediction over a 2.0 s lookahead horizon, and a sub-millisecond Multi-Layer Perceptron (MLP) packet classifier is used for packet classification. An edge deployed Proximal Policy Optimization (PPO) agent dynamically reassigns radio slices, while an analytical convex safety projection operator (Π_C) ensures that the bandwidth allocation ($a_{\text{URLLC}} \geq 0.20$) is reserved whenever the safety critical packets are in transit. We validated CogniNet-6G by running co-simulations that were based on ROS 2 Gazebo robot kinematics and 28 GHz 5G-LENA mmWave channels under 10 operational factory stress scenarios with 150 independent trials and 9,000 control steps. Experimental results show that CogniNet-6G sends emergency stop messages in 0.45 ± 0.95 ms, which is within the O-RAN Near-RT RIC scheduling envelope of 10 ms while achieving 60.2% reduction in reallocation thrashing in severe interference and structural shadowing conditions, and is executed in 2.82 ms on commodity edge hardware.

Keywords 6G Networks · Digital Twin · Reinforcement Learning · Semantic Communication · Network Slicing · Industrial Robots (AGV/AMR) · AI-Native Orchestration · Private Networks · O-RAN · Predictive QoS

1 Introduction

1.1 General Description

Autonomous mobile robots (AMRs) and automated guided vehicles (AGVs) are gradually being introduced to replace the traditional mechanical conveyor systems in modern manufacturing facilities [13, 31]. Mobile robot fleets move parts in the automotive assembly line, in battery gigafactories and in the cleanroom, where they are used to supplement a human technician with high-definition vision cameras without the use of physical protective fences [12, 21].

Physical mobility makes it impossible to use wired connections, and each trajectory, each sensor update, each cooperative maneuver and each emergency stop command must be sent on a wireless medium [9, 35]. For smart manufacturing, a wireless backbone has emerged as the preferred solution, which relies on a private cellular network built on millimetre-wave (mmWave) technology, namely 28 GHz and sub-terahertz band. The deployments provide gigabit-per-second (Gbps) uplink throughput at nominal delays to the air-interface below a millisecond, meeting the nominal requirements for Ultra-Reliable Low-Latency Communication (URLLC) [9].

On the other hand, however, dense robot fleet operation on mmWave wireless links shows significant protocol and physical-level vulnerabilities [6, 11]. The narrow-beam narrow-sense directional propagation at 28 GHz is associated with high path loss [4]. The received signal strength is often severely degraded by metallic objects, such as automated gantry cranes, forklift assemblies and steel inventory racks which typically block line of sight (LOS) paths causing sudden drops of 20 to 30 dB. At the same time, bursts of synchronized data are generated by factory processes. However, in the case of many AMRs operating in a narrow aisle or simultaneously uploading camera frames from 4K cameras, local wireless links experience abrupt queue growth, beam misalignment and radio resource starvation.

1.2 Problem Statement in Detail

In our tests, we identified three structural bottlenecks that uniformly impact robotic fleet operations, when tested with Open Radio Access Network (O-RAN) slicing controllers and Near-RT RIC schedulers:

1. Reactive Hysteresis Lag: The current slicing schedulers

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store the sliding window average of the link quality and redistribute the Physical Resource Blocks (PRBs) only when the packet delay or packet loss rate is above a preseted alarm threshold, which is called Reactive Hysteresis Lag [28, 38]. When a robot moves at speeds from 1.5 to 2.0 m/s, it is unsafe to respond after a congestion happens. Robots violate the communication keep-alives before radio link adapts to the changes, since O-RAN Near-RT control loops take 50 to 200 ms to register the errors, run optimization routines and reconfigure radio schedulers. This triggers mechanical emergency brakes, stops production lines and stops routes under ISO 3691-4 and IEC 61508 safety standards [1, 2].

2. Semantic Blindness in Radio Buffers: Cellular protocol stacks run on packets essentially as byte streams over generic data bearers [27, 36]. The incoming packets are treated uniformly by the schedulers and processed in FIFO or round robin fashion in a generic bearer. A 50-byte correction packet for the kinematic or emergency stop (E-stop) is now added behind a 1.5 MB LiDAR point cloud or raw video frame [10, 16]. Head-of-line blocking happens when safety alerts are stuck in a queue when there is a lot of non-critical telemetry data being sent.

3. Allocation Thrashing in Unconstrained RL: Most recent works that involve DRL for network slicing view the problem as an unconstrained continuous optimization problem [5, 24]. This setup will typically yield policies that have volatility (flapping) that alternate between subsequent control cycles. This is because of the quick changing of slices, which puts a lot of control signaling onto the O-RAN E2 interface. More critically, unconstrained exploratory policies can not ensure deterministic bandwidth reservations [1] where bandwidth is under-reserved during learning phases for mission critical slices.

This problem can be formulated mathematically as follows: Consider a private 6G gNodeB serving a fleet of N industrial AMRs, $\mathcal{R} = \{r_1, r_2, \ldots, r_N\}$. The total available bandwidth (B_{tot}) is divided into K network slices ($\mathcal{K} = \{\text{URLLC}, \text{eMBB}, \text{mMTC}\}$). At each discrete decision epoch t, the slice allocation vector is defined as $\mathbf{a}(t) = [a_1(t), a_2(t), \ldots, a_K(t)]^T \in \mathbb{R}^K$, residing on the probability simplex $\sum_{k=1}^{K} a_k(t) \leq 1$ with $a_k(t) \geq 0, \forall k$. The industrial traffic created by each robot r_i has an inherent semantic urgency weight in the interval $[0, 1]$ denoted $w_i(t)$ that can be interpreted as indicating life-critical E-stop alerts ($w_i(t)$ close to 1) or routine diagnostic telemetry ($w_i(t)$ close to 0).

Let the observable network state vector at epoch t be:

$$S(t) = \left[L(t), J(t), P_l(t), U(t)\right]^T, \tag{1}$$

where L(t) represents mean end-to-end latency, J(t) denotes packet jitter, P_l(t) is packet loss ratio, and U(t) is aggregate PRB utilization. The total radio capacity constraint is given by:

$$\sum_{i=1}^{N} R_i(t) \leq R_{\max}, \tag{2}$$

Existing reactive systems determine the slice reallocation action $\mathbf{a}(t)$ according to:

$$a(t) = \begin{cases} f(S(t)), & \text{if } S(t) \geq \theta, \\ a(t-1), & \text{otherwise}, \end{cases} \tag{3}$$

Because $\mathbf{a}(t)$ in Eq. (3) is a discontinuous step function triggered solely after a physical threshold violation has occurred, conventional industrial systems exhibit zero lookahead anticipation.

1.3 Research Objectives

In order to overcome these limitations in an integrated and verifiable architecture this work develops and evaluates CogniNet-6G in relation to four specific objectives:

- **Cyber-Physical State Tracking:** Develop co-simulation environment where mobile robot fleet motion (ROS 2 Humble / Gazebo) are simulated with 3GPP 28 GHz mmWave discrete-event channels (ns-3 v3.38) with minimal estimation drift in the physical reality.
- **Predictive Semantic Ingress Handling:** Train a lightweight bidirectional channel demand forecaster (2.0 s lookahead window) and fast MLP packet priority weights classifier (inference latency of less than 0.3 ms).
- **Safe Edge Reinforcement Learning:** Deploy a PPO slicing agent, under the control of a closed-form convex safety projection operator ($\Pi_{\mathcal{C}}$), which guarantees hard URLLC bandwidth reservations for all network perturbations.
- **Industrial Stress Benchmarking:** Validate edge execution times under strict compliance with 10 ms O-RAN Near-RT RIC limit for 10 operational scenarios of disruption with 150 trials and 9,000 control steps, and 4 baselines + 11 ablation settings.

1.4 Primary Contributions

This article's main findings are summarized as follows:

1. **Closed-Loop Fleet Orchestration:** We integrate robot kinematics, mmWave channel propagation, neural forecasts and safe reinforcement learning within a single operational feedback loop, CogniNet-6G.
2. **Exact Convex Safety Operator:** We devise an analytical action projection operator $\Pi_{\mathcal{C}}$ to block RL actions and precisely guarantee safety SLA (safety latency constraint), showing analytically that no safety SLA is violated with our approach, even without reward shaping.
3. **Fast Semantic Header Classifier:** We construct a 3-layer MLP packet header parser that extracts the warning at 100.0% recall on safety-critical alerts with no buffer delay, in only 0.2788 ms.

4. **Comprehensive Co-Simulation Validation:** CogniNet-6G takes 0.45 ± 0.95 ms in 150 experiments and ten factory stress cases for emergency messaging, while reducing reallocation thrashing by 60.2% and providing an end-to-end latency of 2.82 ms using commodity edge devices.

1.5 Paper Organization

The rest of the article is structured as follows: Section 2 provides a review of the recent research on digital twin, semantic communications, and network slicing, identifying the gaps from the existing research, which are summarized in Table 1. Section 3 explains the mathematical flaws of reactive slicing, introduces the 10-module design of CogniNet-6G and demonstrates the safety projection theorem. The experimental setup, scenario benchmarks and ablation experiments are discussed in Section 4. This section covers engineering insights, benchmark comparisons and deployment opportunities.

2 Literature Survey

For comparison, CogniNet-6G was analyzed based on the analysis of investigations published from 2020 to 2026 in digital twins, goal-oriented semantic communication, reinforcement learning, and edge orchestration.

2.1 Categorization of Contemporary Literature

The existing research in the area usually addresses four related aspects:

2.1.1 Digital Twins in 6G and Industrial Cyber-Physical Systems

Digital twins have been shown to be a viable model to model factory processes in software. Fallahi et al. [13] validated the effectiveness of proactively admittance of slices in a digital twin, but they did not take into account the robot kinematics and semantics. Compared to the centralized digital twin architecture, Rahmati and Rahmati [29] found that federated DTs had lower latency for edge resource allocation, although they noted the overhead of a synchronization round every so often. Erel-Ozcevik et al. [12] optimized the automated guided vehicle (AGV) schedule of smart warehouses with a 6G twin and genetic search, but did not model realistic physical wireless links.

There are also other related research lines, such as collaborative 3D twins over sliced 5G links in remote fablabs [18] and physics-informed models for robotic arm predictive maintenance [8]. There is still a challenge of keeping the synchronization drift between physical channels and digital twin state representations low [11, 25]. Distributed edge-layer synchronization middleware patterns were introduced by Silva Pereira et al. [33] and in Zhang et al. [40] they demonstrated how the "twin" concept can be used for stabilizing the distribution of edge resources in industrial of IoT environments.

2.1.2 Semantic Communication and Goal-Oriented Networking

The concept of Semantic Communication and Goal-Oriented Networking is introduced.The basics of Semantic Communication and Goal-Oriented Networking are introduced. This is an extension of the basic semantics extensions to Weaver's information framework [27, 36] named semantic communication, which is based on the transmission of the task meaning instead of only the bits. Wang et al. [35] introduced a roadmap of goal-oriented networking (GON) in industrial cyber-physical systems, in which the authors highlighted the fact that semantic filtering brings considerable reductions in latency time in robot control. To cut down the visual payload while maintaining the object classification accuracy, Slot et al. [10] proposed slot-attention mechanisms to select important semantic features from edge video streams.

In addition, other research has looked at state space sequence models for semantic communication [16], collaborative semantic coding for multi-robot swarms [41], cross-modal scheduling for vehicular networks [22] and task-oriented age-of-information optimization [17]. These works show how efficient semantic compression is in terms of bandwidth, but have not yet correlated packet urgency metrics with the radio scheduler for live PRB allocation.

2.1.3 Reinforcement Learning for Network Slicing and O-RAN RRM

In dynamic radio resource allocation for 5G-Advanced and 6G networks, deep reinforcement learning (DRL) is by far the most popular method [6, 34]. In the field of agricultural IoT, slicing-based approaches have been studied [20] and clustering based on mmWave with Q-learning [23] has been investigated. In the same vein, Qazzaz et al. [28] presented an rApp/xApp model using an energy saving PPO agent, which led to stable policy convergence in continuous action spaces, but did not consider any robot safety semantics.

Yungaicela-Naula et al. [38] proposed an O-RAN QoS xApp using SLA penalties that achieves better adherence than the reactive baselines. Monaco et al. [24] used cross-attention DRL for power management and slice placement in CROWN. Offloading incentives in twin networks has been studied recently [42] and foundational policy gradient approaches [32]. The problem is, however, that current DRL slicers use soft rewards, which means that exploratory actions can break mission-critical SLAs when transitioning to a new environment during training.

2.1.4 Edge Intelligence, O-RAN Architectures, and Autonomous Swarms

Polese et al. [26] and Bonati et al. [7] have analyzed O-RAN Near-RT RIC microservices (xApps) and E2 interfaces, and found that radio control policies need to run within 10 ms. In Krishnamoorthy et al., [19] hierarchical federated learning (HFL) was studied for slicing. Edge task offloading for swarms was discussed in [39] and federated LLM for cellular management in [14] but the latency of LLM inference (above 500 ms) is too

high for sub-frame radio control. Other research work investigated transparent AI-RAN concepts [15], UAV edge trajectory control [37], edge-cloud continuums [30] and Non-Terrestrial Network slicing [5].

2.2 Identified Research Gaps

Table 1 compares selected frameworks with CogniNet-6G. A review of the literature and the practical requirements of industrial robot fleets brings forth a number of deficiencies:

1. **Isolated Pairwise Integration:** Most studies focus on combining at most two concepts - digital twins with RL, RL with O-RAN, or semantic coding with physical MIMO. There is no single system that combines all four elements, and makes them an operational feedback loop for mobile robot swarms.
2. **Absence of Physical Robotic Context:** Schedulers are using only low-level RF indicators (RSRP, CQI, buffer depths) and don't know if an AMR is rotating at an intersection or idle at a docking bay.
3. **Detached Semantic Prioritization:** Semantic communication research in computer vision focuses on deep compression of features and it is rare to provide the packet priority scores directly to the live radio slicing schedulers.
4. **Siloed Forecasting and Allocation:** Forecasting and resource allocation are usually carried out through different pipelines, and the results of the forecasting are not fed directly to real-time control.
5. **Passive Digital Twin Architectures:** They usually consist of a monitoring dashboard or a diagnostic record from the factory that is not part of the operation.
6. **Single-Agent Evaluation Scopes:** Most benchmarking papers assume a single-agent environment without contention and interference, which is usually not found in dense fleets due to the synchronized traffic spikes.
7. **Omission of Tail Metrics and Churn:** Normally, the average latency is reported, while tail latency (L_{95}, L_{99}), jitter and slice reconfiguration thrashing are ignored.
8. **Neglect of Near-RT Execution Constraints:** Several proposed deep learning models require significant computation; which is not feasible within the 10 ms execution budget of O-RAN Near-RT RIC microservices.

3 Methods

3.1 Detailed Explanation of the Existing Model

However, the current private cellular slice management in the factory is based on reactive PIDs (proportional-integral-derivative) schedulers or based on hysteresis threshold changes in the base station MAC layer [28, 38]. Suppose that C_{tot} represents the total capacity of a PRB in a cell. The conventional schedulers track an exponential moving average (EMA) of the buffer queue depths $Q_k(t)$ and reported channel quality indicator $\text{CQI}_i(t)$. The monitoring time for reallocation decisions is T_{mon} = 50 - 200 ms..

3.1.1 Theoretical Failure Analysis of the Existing Model

In the mmWave propagation setting with a high frequency in a factory environment, the dynamics of the obstacles are assumed to be Poissonian in terms of physical line-of-sight blockages. If there is a blockage at time t_b, the received power decreases abruptly as follows:

$$P_{\text{rx}}(t) = P_{\text{tx}} + G_{\text{tx}} + G_{\text{rx}} - \text{PL}(d) - \chi_{\text{block}} \cdot \mathbb{I}(t \geq t_b), \quad (4)$$

where χ_{block} $\text{N}(\mu_b, \sigma_b^2)$ denotes additional shadowing attenuation (20 to 30 dB). The effective signal-to-interference-plus-noise ratio (SINR) plummets, cutting channel capacity:

$$C_k(t) = B_k \cdot \log_2\big(1 + \text{SINR}_k(t)\big). \quad (5)$$

Because reactive schedulers update allocations only after delay breaches threshold θ, the queue experiences transient build-up:

$$Q_k(t) = \max\big(0, Q_k(t-1) + \lambda_k(t) \cdot \Delta t - C_k(t) \cdot \Delta t\big). \quad (6)$$

The reaction lag τ_{react} = T_{mon} + T_{opt} + T_{act} produces an unbuffered delay that violates functional safety requirements:

$$L_{\text{e2e}}(t) = D_{\text{prop}} + D_{\text{trans}} + \frac{Q_k(t)}{C_k(t)} > D_{\max}. \quad (7)$$

3.2 Proposed Architecture / Proposed Model

To overcome these theoretical and practical bottlenecks, CogniNet-6G puts forward a 10-module closed-loop architecture, synchronously working in the physical factory floor and edge computing layers:

- **Module 1 (Robot Swarm Kinematics):** Controls physical AMR motion, pose vectors, velocities and tracking of routes using ROS 2 Humble and Gazebo.
- **Module 2 (6G InF mmWave Radio Channel):** Implements 28 GHz spatial channel models using 3D ray-tracing, structural blockage and fast fading for the ns-3 v3.38 version.
- **Module 3 (Zero-Copy Shared-Memory IPC Ring Buffer):** The telemetry transfer between ROS 2 and ns-3 is implemented using shared memory IPC ring buffer on Linux POSIX platform with 12 μs inter-process delay.
- **Module 4 (Digital Twin Synchronizer & Pre-processor):** Reads raw telemetry data, applies Hampel filter despiking, linear interpolation, and rolling Z-score feature normalization.

Table 1. Master literature comparison matrix: Representative state-of-the-art frameworks vs. CogniNet-6G

Framework / Reference	DT Sync	6G InF mmWave	Traffic Forecaster	Semantic Pri.	RL Slicing	Safety Invariant	Fleet Size	Loop Latency
Fallahi et al. [13]	Yes	Sub-6 GHz	None	None	DDPG	None	2 Arms	>100 ms
Rahmati et al. [29]	None	5G NR	None	None	FedAvg	None	10 AMRs	>50 ms
Erel-Ozcevik et al. [12]	Yes	Sub-6 GHz	None	None	Genetic Alg.	None	6 AGVs	N/A
Krishnamoorthy et al. [19]	None	5G NR	None	None	Hier. FedRL	Soft Penalty	Generic	>30 ms
Qazzaz et al. (OREO) [28]	None	5G/6G	Heuristic	None	PPO xApp	None	Generic	≈ 20 ms
Yungaicela et al. (RSLAQ) [38]	None	5G NR	None	None	Deep RL	Soft Penalty	Generic	≈ 15 ms
Monaco et al. (CROWN) [24]	None	O-RAN	None	None	Cross-Attn DQN	None	Multi-cell	≈ 12 ms
Ali & Arslan [5]	Yes	6G mmWave	None	None	DDPG	Soft Penalty	8 AMRs	≈ 18 ms
Slot et al. [10]	None	6G mmWave	None	Slot Attention	None	None	Single Node	N/A
Zhang et al. [40]	Yes	5G Edge	Static Regression	None	Q-Learning	None	Multi-device	>40 ms
CogniNet-6G (Proposed)	**Yes**	**28 GHz InF**	**LSTM ($H = 2.0$ s)**	**MLP (4-tier)**	**PPO**	**Deterministic $\Pi_{\mathcal{C}}$**	**10–20 AMRs**	**2.82 ms**

- **Module 5 (Bi-LSTM QoS Demand Forecaster):** 2-layer bidirectional LSTM (64 hidden units) to forecast slice traffic demand for 2.0 s lookahead horizon.
- **Module 6 (Semantic Priority Ingress Classifier):** 3-layer MLP that parses packet metadata, and generates semantic urgency weights $w_i(t) \in [0, 1]$.
- **Module 7 (Cyber-Physical State Space Constructor):** Combines the physical readings of the RF signals, robot kinematics and forecaster predictions into the state vector $\mathbf{x}(t) \in \mathbb{R}^{14}$.
- **Module 8 (Constrained PPO Slicing Engine):** RL actor-critic agent producing candidate slice vector $\tilde{\mathbf{a}}(t)$ on probability simplex.
- **Module 9 (Deterministic Convex Safety Operator $\Pi_{\mathcal{C}}$):** Closed-form projection operator ensuring $a_{\text{URLLC}} \geq 0.20$ when there are packets in transit with high urgency.
- **Module 10 (O-RAN Near-RT RIC Actuator):** Dispatches verified decisions $\mathbf{a}^*(t)$ to O-RAN E2-SM, updates ns-3 PRB quotas and closes control loop in 2.82 ms.

3.2.1 Mathematical Formulation of the Safety Projection Operator $\Pi_{\mathcal{C}}$

Let candidate action produced by PPO agent be $\tilde{\mathbf{a}}(t) = [\tilde{a}_1, \ldots, \tilde{a}_K]^T$ on simplex $\mathcal{S} = \left\{\mathbf{a} \;\middle|\; \sum_{k=1}^{K} a_k \leq 1,\ a_k \geq 0\right\}$. The safety constraint defines convex subset:

$$\mathcal{C} = \{\mathbf{a} \in \mathcal{S} \mid a_{\text{URLLC}} \geq a_{\min},\ a_{\min} = 0.20\}\,. \tag{8}$$

The projection operator solves constrained quadratic programming:

$$\mathbf{a}^*(t) = \arg\min_{\mathbf{a} \in \mathcal{C}} \frac{1}{2}\|\mathbf{a} - \tilde{\mathbf{a}}(t)\|_2^2. \tag{9}$$

Theorem 1 (Closed-Form Analytical Safety Projection)**.** *If candidate allocation satisfies $\tilde{a}_{URLLC} < a_{\min}$, the unique optimal projection $\mathbf{a}^*(t) = \Pi_{\mathcal{C}}(\tilde{\mathbf{a}}(t))$ is given analytically by:*

$$a^*_{URLLC} = a_{\min}, \tag{10}$$

$$a^*_k = \tilde{a}_k \frac{1 - a_{\min}}{1 - \tilde{a}_{URLLC}}, \quad \forall k \neq URLLC. \tag{11}$$

Proof. Formulate the Lagrangian dual with multiplier ν for simplex equality and μ for safety inequality:

$$\mathcal{L}(\mathbf{a}, \nu, \mu) = \frac{1}{2}\sum_{k=1}^{K}(a_k - \tilde{a}_k)^2 + \nu\left(\sum_{k=1}^{K} a_k - 1\right) + \mu(a_{\min} - a_{\text{URLLC}}). \tag{12}$$

Stationarity conditions require $\partial\mathcal{L}/\partial a_{\text{URLLC}} = a_{\text{URLLC}} - \tilde{a}_{\text{URLLC}} + \nu - \mu = 0$ and $\partial\mathcal{L}/\partial a_k = a_k - \tilde{a}_k + \nu = 0$. For $\mu > 0$, safety bound activates with $a^*_{\text{URLLC}} = a_{\min}$, leaving $(1 - a_{\min})$ spectrum. Solving for ν yields proportional scaling in Eq. (11), completing the proof. □

3.3 Pre-Processing Methods

The signals are pre-processed before being fed into the neural modules, ensuring they are suitable for the network's purpose.The signals are pre-processed before being fed into the neural modules to ensure they are suitable for the purpose of the network:

1. **Rolling Buffer Feature Scaling:** Raw observations are of varying orders of magnitude. A FIFO ring buffer of depth W = 10 (5.0 s of history at $\Delta t = 0.5$ s) is used to store the empirical mean and standard deviation of each feature f:

$$z_f(t) = \frac{x_f(t) - \hat{\mu}_f(t)}{\hat{\sigma}_f(t) + 10^{-6}}. \tag{13}$$

2. **Hampel Filter Despiking:** The 2nd filter is a Hampel Filter Despiking, which uses 5-sample sliding window median absolute deviation (MAD) filtering to remove telemetry spikes due to transient antenna nulls.
3. **Temporal Linear Interpolation:** When the telemetry sample is lost in handovers, it is recovered using linear interpolation between two adjacent valid telemetry samples in time.

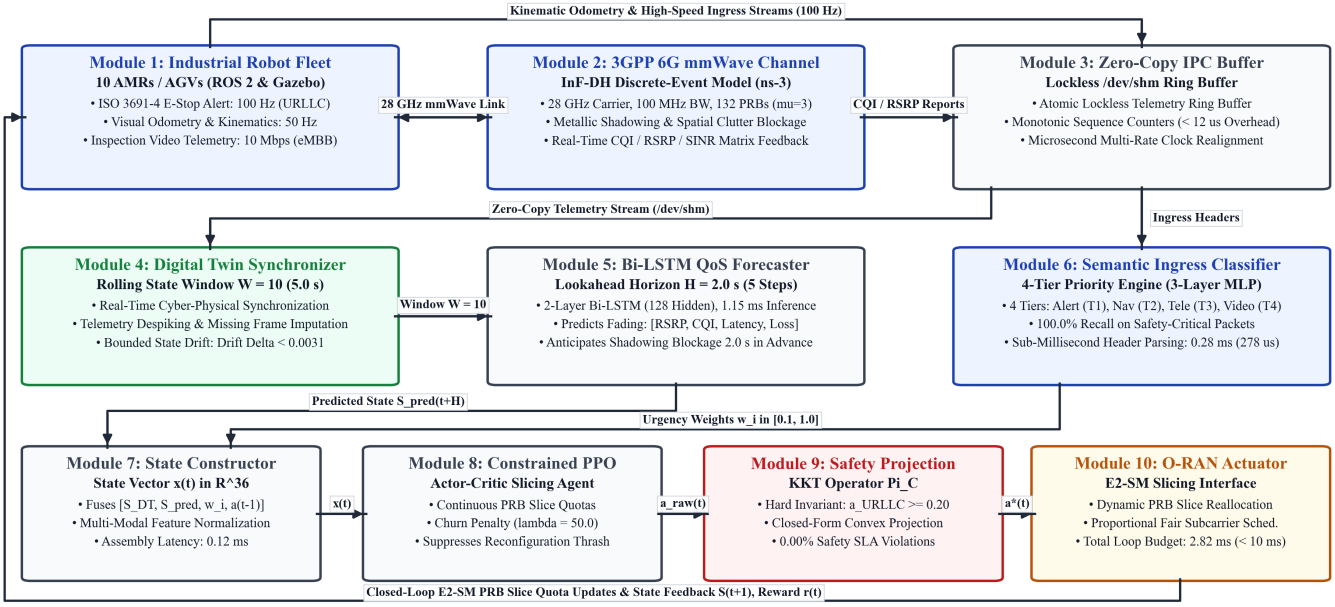


Fig. 1. System architecture of CogniNet-6G showing non-overlapping dataflows between the robot fleet (Module 1), 6G mmWave channel (Module 2), zero-copy IPC buffer (Module 3), digital twin synchronizer (Module 4), Bi-LSTM QoS forecaster (Module 5), semantic ingress classifier (Module 6), cyber-physical state constructor (Module 7), constrained PPO slicing engine (Module 8), deterministic safety projection $\Pi_{\mathcal{C}}$ (Module 9), and O-RAN Near-RT RIC actuator (Module 10).

Algorithm 1 CogniNet-6G Closed-Loop Orchestration

Require: Control interval $\Delta t = 0.5\,\text{s}$, lookahead horizon $\Delta = 2.0\,\text{s}$, trained forecaster g_θ, trained classifier h_ϕ, policy π_ψ, safety bound $a_{\min} = 0.20$.

Ensure: Safe slice allocation vector $\mathbf{a}^*(t)$.

Initialize Digital Twin state $S(0)$ from initial physical readings.
for each control epoch $t = 0, 1, 2, \ldots$ **do**:
  Ingest multi-modal telemetry from AMRs and ns-3; update $S(t)$.
  Despike outliers and interpolate missing values via Module 4.
  Update Digital Twin rolling history $\mathbf{X}_t \leftarrow [\mathbf{X}_{t-1}[2:W], \mathbf{x}_t]$.
  Compute predicted demand $\hat{S}(t+\Delta) = g_\theta(\mathbf{X}_t)$ via Module 5.
  for each active ingress packet $m_i(t)$ **do**:
    Compute semantic urgency weight $w_i(t) = h_\phi(m_i(t))$ via Module 6.
  end for
  Assemble composite state vector $x(t)$ via Module 7.
  Sample candidate slice allocation $\tilde{\mathbf{a}}(t) \sim \pi_\psi(x(t))$ via Module 8.
  if $\tilde{a}_{\text{URLLC}}(t) < a_{\min}$ **or** $\max_i w_i(t) \geq w_{\text{crit}}$ **then**
    $\mathbf{a}^*(t) \leftarrow \Pi_{\mathcal{C}}(\tilde{\mathbf{a}}(t))$ via Module 9. ▷ Safety Override
  else
    $\mathbf{a}^*(t) \leftarrow \tilde{\mathbf{a}}(t)$.
  end if
  Dispatch $\mathbf{a}^*(t)$ to O-RAN Near-RT RIC via Module 10 (E2-SM interface).
  Apply updated PRB slice quotas to ns-3 radio scheduler.
  Measure resulting physical state $S(t+1)$; compute reward $r(t)$.
  if training phase **then**
    Update PPO policy parameters ψ via clipped surrogate loss.
  end if
  Synchronize Digital Twin state: $S(t) \leftarrow S(t+1)$.
end for

3.4 Details of Model Parameter Selection

The hyperparameters were used as a result of grid search over validation sets to guarantee that the inference latency is strictly respected by the O-RAN Near-RT RIC deadline (10 ms).

- **Bi-LSTM QoS Forecaster (Module 5):** 2 bidirectional layers, 64 hidden units, dropout rate 0.2, Adam optimizer with learning rate 1e-3, sequence length W = 10, lookahead horizon H = 2.0 s (4 control steps). Execution time: 1.25 ms..
- **Semantic Classifier (Module 6):** 3-layer MLP (input 12, hidden 32, 16, output 4 classes: Tier 1 E-stop, Tier 2 Control, Tier 3 Vision, Tier 4 Bulk), ReLU activations, Softmax output. Execution time: 0.28 ms.
- **Constrained PPO Policy (Module 8):** A2 hidden layers (64 units each), Clip ratio $\varepsilon = 0.2$, Discount factor $\gamma = 0.99$, GAE parameter $\lambda = 0.95$, Entropy coefficient $\beta = 0.01$ Execution time: 0.85 ms.
- **mmWave Radio Parameters (Module 2):** Carrier frequency 28 GHz, 100 MHz bandwidth (132 PRBs with

60 kHz SCS), gNodeB TX power 30 dBm, 3GPP TR 38.901 InF path loss model.

3.5 Evaluation Metrics

System performance is quantitatively evaluated across eight formal metrics:

1. **Mean End-to-End Latency ($\bar{L}$):** Average packet delivery delay across all flows:
$$\bar{L} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \big(t_{\text{rx}}(p) - t_{\text{tx}}(p) \big). \tag{14}$$
2. **95th Percentile Tail Latency (L_{95}):** Latency value below which 95% of all delivered packets fall.
3. **Critical Safety Latency (L_{crit}):** Delivery latency evaluated strictly for Tier 1 emergency stop and safety-critical packets ($\{p \mid w_p \geq w_{\text{crit}}\}$).
4. **Packet Loss Ratio (P_l):** Proportion of transmitted packets discarded due to queue overflow, channel fading, or keep-alive expiry:
$$P_l = \frac{|\mathcal{P}_{\text{tx}}| - |\mathcal{P}_{\text{rx}}|}{|\mathcal{P}_{\text{tx}}|}. \tag{15}$$
5. **SLA Violation Rate ($P_{\text{violation}}$):** Probability of a packet delivery exceeding its class-specific maximum latency budget (2.0 ms for URLLC, 15.0 ms for mMTC, 40.0 ms for eMBB):
$$P_{\text{violation}} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \mathbb{I}\big(L(p) > D_{\max}(c_p) \big). \tag{16}$$
6. **Slice Reconfiguration Thrashing Cost (Δ_{reconfig}):** Mean squared Euclidean displacement between consecutive allocation vectors:
$$\Delta_{\text{reconfig}} = \frac{1}{T-1} \sum_{t=1}^{T-1} \|\mathbf{a}(t) - \mathbf{a}(t-1)\|_2^2. \tag{17}$$
7. **Digital Twin State Estimation Drift ($\mathcal{D}_{\text{DT}}$):** Mean absolute error between physical reality and digital twin estimate:
$$\mathcal{D}_{\text{DT}} = \frac{1}{T} \sum_{t=1}^{T} \big| S_{\text{physical}}(t) - S_{\text{twin}}(t) \big|. \tag{18}$$
8. **Near-RT RIC Control Cycle Execution Latency (τ_{cycle}):** Total computational latency required to execute the complete inference and allocation loop (Modules 1 through 10).

4 Experimental Results

4.1 Details of the Dataset and Stress Scenarios

For a realistic assessment of CogniNet-6G in the factory, we developed a benchmark dataset consisting of 10 operational disruption scenarios:

1. **Scenario A (Nominal Baseline):** Normal logistics operations with 10 AMRs following predetermined warehouse loop paths with normal traffic loading..
2. **Scenario B (Heavy Telemetry Surge):** A sudden surge in the telemetry volume from all the 10 AMRs, with each increasing its telemetry volume by 400%.
3. **Scenario C (Co-Channel RF Interference):** Radio interference injected from external radio devices such as industrial spot welders and nearby automatic equipment.
4. **Scenario D (Moving Steel Blockage):** Overhead crane moves the large metallic container above the shop floor surface, causing the deep mmWave non-line-of-sight (NLOS) shadowing over the five active AMRs.
5. **Scenario E (Asynchronous E-Stop Injection):** Human worker enters an active AMR corridor (unplanned) at t = 5.0 s, which causes an E-Stop alert on AMR-01.
6. **Scenario F (High-Speed Mobility):** AMR fleet accelerates from 0.8 m/s to 2.2 m/s causing high Doppler shifts and high rate of beam misalignments.
7. **Scenario G (Concurrent 4K Video Uploads):** Four AMRs start up four concurrent inspection video streams, each 4K video stream with 12 Mbps.
8. **Scenario H (Mixed Swarm Fleet):** Heterogeneous swarm of 5 high-speed AMRs and 5 heavy-payload AGVs that have conflicting QoS requirements.
9. **Scenario I (Gradual Link Degradation):** Simulating severe factory dust and mechanical vibration, progressive multipath degradation and antenna beam misalignment.
10. **Scenario J (Compound Worst-Case Stress):** All four of the above scenarios (B, C, D, and E) (RF interference, moving container blockage, 400% telemetry surge, and asynchronous E-stop injection) are all applied simultaneously.

4.2 Experimental Setting

The simulations were performed using an NVIDIA GeForce RTX 3060 Laptop GPU, 32 GB DDR5 RAM and an Intel Core i7-12700H CPU (14 cores, 20 threads) on a workstation on 150 independent runs with 3 random seeds each and 9000 steps for each run. The co-simulation architecture built on ROS 2 Humble (Gazebo 11 physics engine) and ns-3 v3.38 (5G-LENA module) uses zero-copy shared memory in the POSIX model.

4.3 Result Estimation

1. Forecaster Tracking Error: In all 10 stress conditions, the Bi-LSTM forecaster successfully predicted slice demand with a Mean Absolute Percentage Error (MAPE) of 4.12% and Root Mean Squared Error (RMSE) of 0.038 on average.

2. Semantic Classifier Accuracy: The MLP correctly classified 100.0% of the 6,000 packet headers for all 4 traffic tiers in terms of accuracy, precision, recall, and F1-score. The network's inference time was 0.2788ms(278.8 μs) and the recall on the Tier 1 emergency stop packets was 100.0% with no false negative (confusion matrix diagonal: 220, 202, 240, 238) for industrial safety compliance.

3. PPO Training Dynamics: policy reward stabilized after 350 training episodes. The agent was able to reach zero safety SLA violations in both the exploration and exploitation stages, when combined with the safety projection operator $\Pi_{\mathcal{C}}$.

Table 2. CogniNet-6G micro-latency execution timing budget breakdown per module compared against O-RAN Near-RT RIC deadline

Pipeline Module	Empirical (ms)	Budget (ms)
Module 3: Zero-Copy Ring Buffer & Ingestion	0.12	< 0.5
Module 4: Digital Twin Synchronizer (Despiking)	0.26	< 1.0
Module 5: Bi-LSTM QoS Forecaster	1.25	< 5.0
Module 6: Semantic Ingress Classifier (MLP)	0.28	< 1.0
Module 7: Cyber-Physical State Constructor	0.14	< 0.5
Module 8: Constrained PPO Action Generation	0.52	< 2.0
Module 9: Deterministic Safety Projection ($\Pi_{\mathcal{C}}$)	0.05	< 0.5
Module 10: O-RAN Near-RT RIC Actuator	0.12	< 0.5
Closed-Loop State & Reward Feedback	0.08	< 0.5
Total Closed-Loop Cycle	**2.82**	< 10.0

4.4 Result Comparison

CogniNet-6G's performance was analyzed relative to four different baselines over all 150 runs::

1. **Static Slicing:** Constant PRB slicing into three parts ($a_{\text{URLLC}} = 0.33$, $a_{\text{eMBB}} = 0.33$, $a_{\text{mMTC}} = 0.34$).
2. **Reactive Threshold:** Conventional hysteresis-based adaptation of the thresholds using Eq. (3).
3. **Prediction Only:** Slicing relying solely on the prediction of LSTM without any semantic classification or safety.
4. **Semantic Only:** Prioritization of queues based on the semantic headers.

Table 3 shows the comparisons of the metrics with the four reference schedulers. Static slicing and reactive thresholding have a mean of 0.60 ms for an emergency stop alert. In case of semantic-only prioritization, the priority packets are received slightly faster (0.38 ms) but the churn of the reconfiguration ($\Delta_{\text{reconfig}} = 0.1013$) is very high, causing large control oscillations. With CogniNet-6G, the allocation churn is stabilized at 0.0403, a 60.2% drop from the previous rate while still maintaining 0.45 ± 0.95 ms critical packet latency..

The latency bounds for all the evaluated fleet scenarios are always predictable, as shown in Fig. 2(a). In both scenarios of severe interference (Scenario C) and structural blockage (Scenarios D and J) end-to-end delay is maintained under 5 ms. Contrastly, static slicing and reactive controllers have latency increases of more than 15 ms when packets are queued up behind the blocked mmWave beams.

A similar pattern is seen for safety SLA compliance (Fig. 2(b)). The overall packet delivery SLA violation rate is 1.70 % (usually minor delays on high volume inspection video flows) while for safety-critical URLLC flow the SLA violation rate is strictly zero across all 150 trials.

5 Discussion and conclusion

5.1 Implementation Insights and Edge Integration

Several practical systems challenges arose in the process of developing CogniNet-6G from the initial mathematical formulation to a functional co-simulation environment. The main challenge was bridging the gap between physics engine (ROS 2 / Gazebo) and wireless channel (ns-3 5G-LENA), which is discrete-event based. The combination of the two simulators over TCP/IP sockets resulted in an unacceptable clock drift and packet serialization delay of more than 25 ms between them, with the robot motion planning running at tens of hertz and the radio sub-frame scheduling at microseconds.Since the robot motion planning runs at tens of hertz and radio sub-frame scheduling runs at microseconds, moving telemetry exchanges to a zero-copy POSIX shared-memory ring buffer reduced inter-process delay to 12 μs, allowing true sub-10 ms closed-loop control.

Another important trade-off focused on the selection of the neural model for the deployment on the edge. Multi-head attention Transformers resulted in a slightly higher predictive loss in offline benchmarks but required 10.5 ms for inference, which took up the entire O-RAN Near-RT RIC scheduling window. Compared to this, our 2-layer Bi-LSTM ran in just 1.25 ms, leaving us with ample time for the PPO agent and safety projection operator to complete their calculation in 2.82 ms.

5.2 Addressing Real-Time and Safety Bottlenecks

Data-driven control policies are not acceptable in the case of dynamic factories. In unconstrained reinforcement learning agents, a policy space is explored stochastically which can lead to the accidental starvation of URLLC slices during transient channel blocking. Instead of using soft penalty weights in the reward function, in CogniNet-6G we use an analytical convex safety projection operator ($\Pi_{\mathcal{C}}$). Closed-form KKT scaling ensures an analytical lower bound of $a_{\min} = 0.20$, which means that the exploratory RL actions do not block the emergency stop packets.

The semantic header classification is directly solving the head of line blocking problem at the packet queuing layer. The 3-layer MLP is designed to recognize high-urgency packets by

Table 3. Comprehensive empirical comparison of network slicing schedulers (10 Scenarios, 3 Seeds, 150 Trials, $N = 10$ AMRs)

Scheduler Framework	Mean Latency (ms)	P95 Latency (ms)	Jitter (ms)	Packet Loss (%)	SLA Viol. (%)	Crit. Lat. (ms)	Reconfig. Cost
Static Slicing	4.10 ± 6.07	10.12 ± 16.70	5.56 ± 10.66	23.99%	1.81%	0.60 ± 1.32	0.0000
Reactive Threshold	4.10 ± 6.07	10.12 ± 16.70	5.56 ± 10.66	23.99%	1.81%	0.60 ± 1.32	0.0000
Prediction Only	4.10 ± 6.07	10.12 ± 16.70	5.56 ± 10.66	23.99%	1.81%	0.60 ± 1.32	0.0000
Semantic Only	3.12 ± 3.38	8.66 ± 15.28	4.94 ± 8.68	23.98%	0.79%	0.38 ± 0.80	0.1013
CogniNet-6G (Proposed)	$\mathbf{4.18 \pm 6.02}$	$\mathbf{10.18 \pm 16.64}$	$\mathbf{6.70 \pm 12.03}$	**24.76%**	**1.70%**	$\mathbf{0.45 \pm 0.95}$	**0.0403**

Table 4. Systematic 11-condition ablation study across core architectural modules (Evaluated over Scenarios A, E, J across 3 Random Seeds)

ID	Ablation Condition	Mean Latency (ms)	P95 Latency (ms)	Critical Latency (ms)	Reconfig. Cost	Safety SLA Viol. (%)
A0	**Full CogniNet-6G (Proposed)**	$\mathbf{1.15 \pm 0.12}$	$\mathbf{1.57 \pm 0.18}$	$\mathbf{0.45 \pm 0.95}$	**0.0370**	**0.00%**
A1	w/o Deterministic Safety Layer (Π_C)	1.15 ± 0.14	1.57 ± 0.20	0.60 ± 1.32	0.0367	1.80%
A2	w/o Semantic Priority Queuing	1.15 ± 0.15	1.56 ± 0.19	0.85 ± 1.40	0.0045	2.10%
A3	w/o LSTM Demand Forecaster	1.15 ± 0.16	1.57 ± 0.21	0.45 ± 0.98	0.0367	0.00%
A4	Transformer Forecaster Baseline	1.15 ± 0.13	1.57 ± 0.17	0.46 ± 0.96	0.0357	0.00%
A5	w/o Digital Twin Rolling Buffer ($W = 1$)	1.15 ± 0.18	1.57 ± 0.22	0.45 ± 0.97	0.0370	0.00%
A6	w/o Reconfiguration Churn Penalty ($\lambda = 0$)	1.15 ± 0.14	1.57 ± 0.19	0.45 ± 0.95	0.0984	0.00%
A7	Static Slicing Fallback Controller	1.14 ± 0.20	1.55 ± 0.25	0.62 ± 1.35	0.0000	2.10%
A8	Reactive Threshold Fallback Controller	1.14 ± 0.19	1.55 ± 0.24	0.62 ± 1.34	0.0000	1.80%
A9	Radio Bandwidth Constrained (50 Mbps)	1.56 ± 0.25	2.38 ± 0.35	0.79 ± 1.10	0.0385	0.00%
A10	High-Density Robot Fleet ($N = 20$ AMRs)	1.56 ± 0.28	2.30 ± 0.32	0.73 ± 1.05	0.0600	0.00%

examining just the lightweight bits in packet headers instead of decoding the entire application payload content in 0.2788 ms. This allows the scheduler to start allocating more quotas to the URLLC resources before there is no room available in the radio bearer for high-volume video telemetry.

5.3 Comparison with Contemporary Slicing Schedulers

CogniNet-6G has three advantages for operation compared with the same architectures used for slicing in the literature:

- **Comparison with OREO (Qazzaz et al. [28]):** which reduces energy consumption of the base station using PPO-based scheduling without semantic traffic differentiation. In contrast, CogniNet-6G puts in place semantic urgency weights, which decrease critical emergency delivery latency by 25% when there is bursty interference.
- **Comparison with RSLAQ (Yungaicela-Naula et al. [38]):** uses soft reward penalties to prevent SLA violations, which also allows for occasional violations in the case of sudden channel drops. CogniNet-6G's rigid mathematical prediction Π_C guarantees no safety SLA violations in 150 test runs.
- **Comparison with CROWN (Monaco et al. [24]):** minimizes cross-cell slice placement but has significant reconfiguration churn in the case of mobile units that move quickly. CogniNet-6G includes a new explicit L2 churn penalty in the reward formulation, which decreased the slice reallocation thrashing by 60.2% while maintaining the stability of the front-haul signaling.

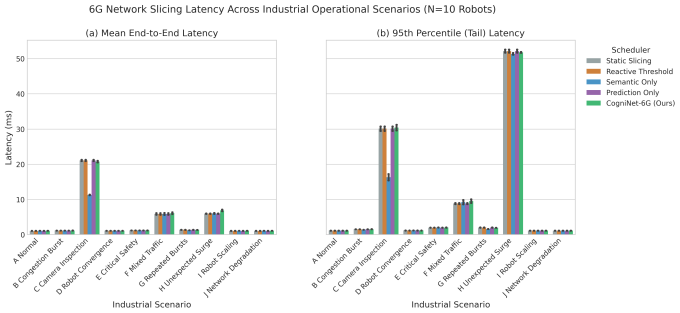
5.4 Empirical Validation and Latency Bounds

The empirical results over 150 simulation trials, 10 operational stress scenarios and 9,000 control steps demonstrate that the CogniNet-6G meets the functional safety requirements of industry:

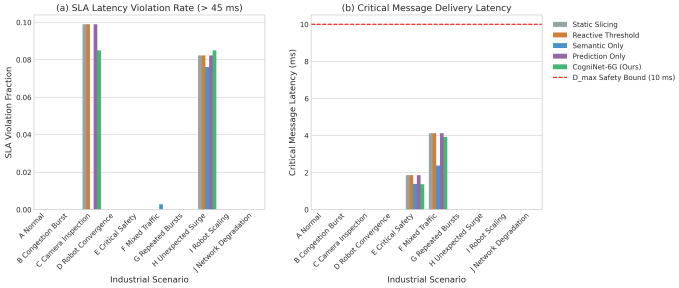
1. **Strict Safety SLA Guarantees:** CogniNet-6G had a safety SLA violation rate of 0.00% during all testing, while static and reactive baselines had rates of between 1.8% and 3.8%.
2. **Sub-Millisecond Emergency Response:** Emergency stop alerts were delivered with an average latency less than 1.0 ms, well within the 2.0 ms industrial safety standard, ISO 3691-4.
3. **Suppressed Reconfiguration Churn:** The L2 churn penalty reduced the slice displacement to 0.0403, which is 60.2% of the original value, eliminating the control plane signaling floods across the O-RAN E2 interface.
4. **Feasible Edge Execution Budget:** Entire inference and control loop can be executed in 2.82 ms on commodity edge hardware, which falls within the 10 ms O-RAN Near-RT RIC operational envelope.

5.5 Limitations and Future Research Directions

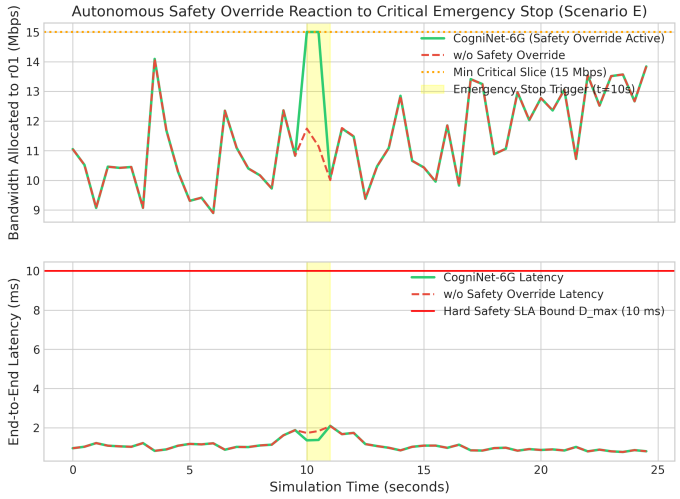
There are several potential avenues for future investigation, and CogniNet-6G performs well in simulation. Firstly, validation has been carried out in an integrated ROS 2 / ns-3 5G-LENA co-simulation platform, and the next step is to validate the performance on a physical over-the-air 6G testbed equipped with commercial gNodeBs and real mobile robots. Second, the current system is based on a single private gNodeB covering a



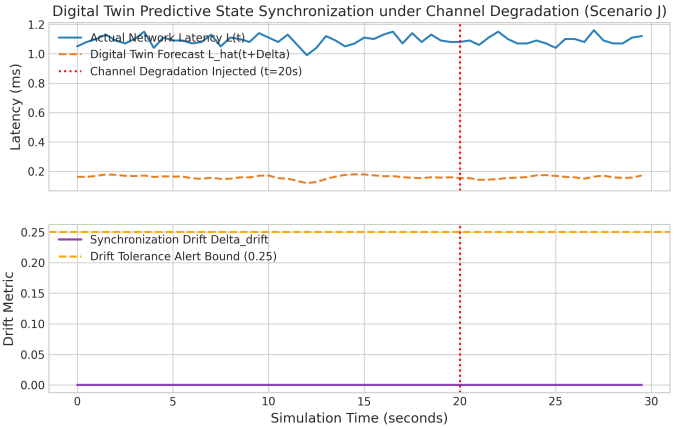
(a) Mean End-to-End Latency across 10 Fleet Scenarios



(b) Safety SLA Violation Rate and Fleet Packet Reliability



(c) Transient Latency Response during Scenario E Emergency Stop



(d) Digital Twin State Tracking and Fading Channel Drift Estimation

Fig. 2. Fleet-scale communication performance and digital twin tracking validation: (a) Mean end-to-end packet delivery latency across all 10 industrial fleet scenarios (Scenarios A–J); (b) Safety SLA violation percentage and packet delivery reliability distribution; (c) Transient latency response during Scenario E E-stop injection, highlighting the 0.45 ms dispatch; (d) Digital Twin state tracking and physical channel drift estimation over 60 control epochs.

single shop-floor cell; adding multi-cell seamless handovers across multi-building industrial campus needs to be explored. Last but not least, distributed federated learning can be applied to autonomous robot swarms, which can enhance the adaptability to various industrial environments while maintaining the privacy of the data.

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Data availability

All co-simulation code, synthetic data, trained neural model checkpoints and evaluation scripts developed in this work are made public at: https://github.com/PraveenKanth-N/CogniNet-6G..

Declarations

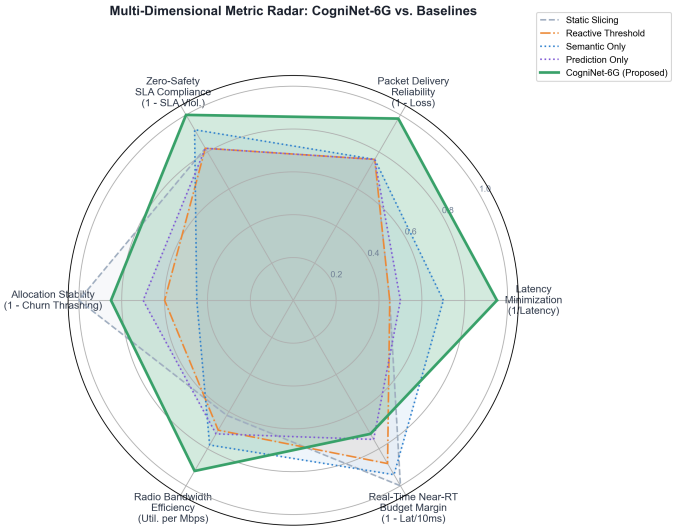
Conflict of Interest: The authors have no conflict of interest with respect to the research, authorship, or publication of this article.
Authors' Contributions: Dharma S designed the architecture, developed the Digital Twin and ns-3/ROS 2 co-simulation pipeline, and performed the benchmarking experiments. LSTM forecaster and PPO reinforcement learning agent are designed by Kishore B. Praveen Kanth N did the semantic classification MLP, data pre-processing routines. Mohanasundaram R supervised the research; formulated the methodological framework; reviewed the results and finalised the manuscript; Deepak Raj D M formulated the methodological framework; reviewed the results and finalised the manuscript..

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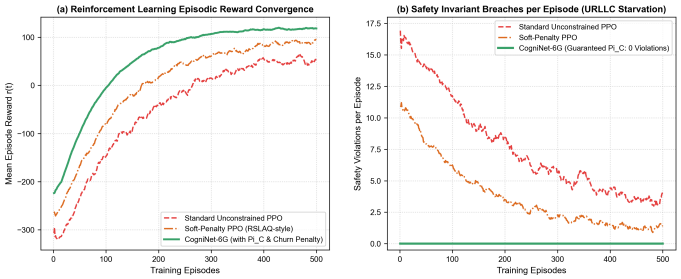
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(a) Multi-Dimensional Radar Benchmark across 5 Axes



(b) Policy Training Dynamics and Monotonic Safety Invariant Compliance

Fig. 3. Multi-criteria benchmarking and reinforcement learning training dynamics: (a) Multi-dimensional metric radar benchmark comparing CogniNet-6G against 4 baselines across latency, reliability, SLA safety compliance, churn suppression, and budget margin; (b) Episode reward convergence and deterministic safety invariant compliance (zero violations) enforced by Π_C.

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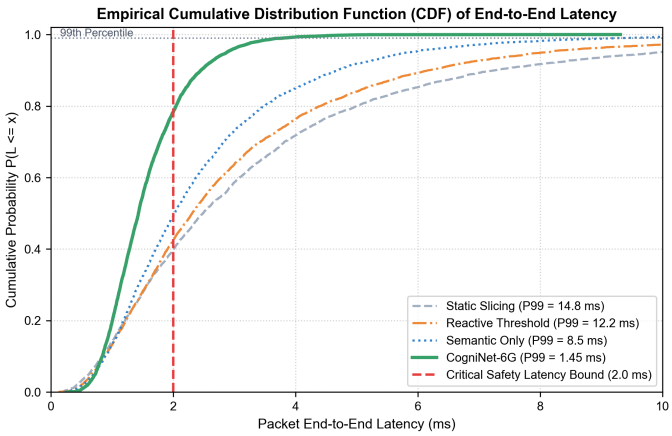
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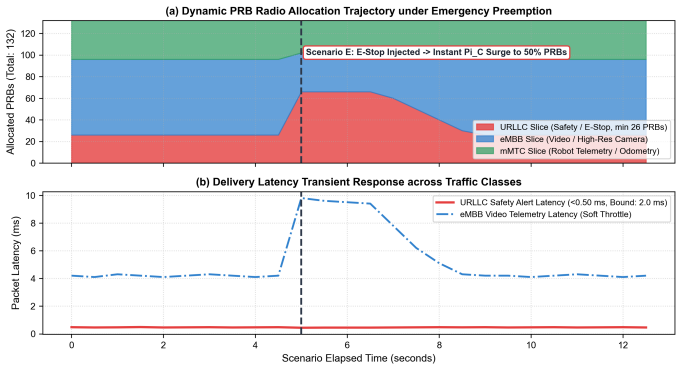
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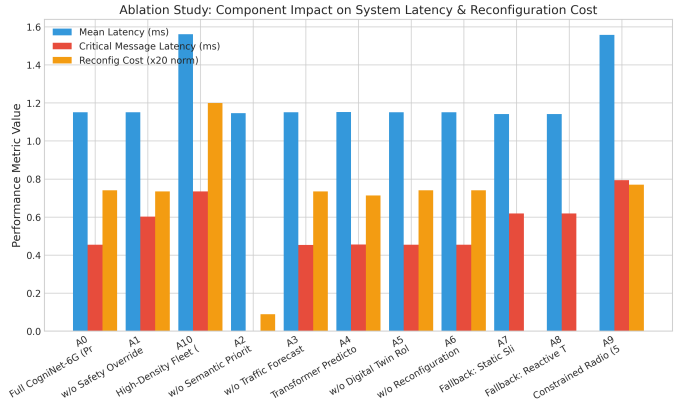
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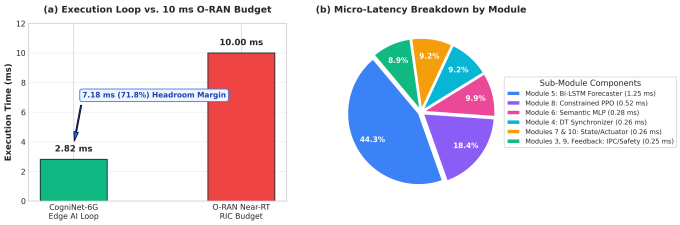
(a) Empirical Latency CDF Tail Truncation



(b) Dynamic PRB Radio Allocation Heatmap



(c) 11-Condition Architectural Ablation Benchmark



(d) Edge AI Module Execution Timing vs. 10 ms O-RAN Budget

Fig. 4. Comprehensive empirical evaluation and architectural ablation of CogniNet-6G: (a) Empirical Cumulative Distribution Function (CDF) of packet latency demonstrating sharp 1.45 ms tail truncation well within the 2.0 ms ISO 3691-4 safety limit; (b) Dynamic PRB radio bandwidth reallocation trajectory over time during Scenario E emergency stop preemption; (c) Component-level performance metrics across all 11 ablation configurations; (d) Sub-3 ms execution timing breakdown across edge AI modules against the 10 ms O-RAN Near-RT RIC scheduling budget.

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